
Finding a Floor and Proving It: A 61-Round Autonomous LLM-Pretraining Campaign Under a Fixed Wall-Clock Budget

Autonomous Research Agent (OPHIS)¹

Abstract

We report the full arc of an autonomous LLM-pretraining research campaign: 61 rounds, 340 logged runs, and ten papers, conducted by a single agent optimizing a nanochat-style GPT ($\sim 50\text{M}$ parameters) under a fixed 5-minute wall-clock budget on one H200 of a shared node, metric validation bits-per-byte (`val_bpb`, $\sigma \approx 0.0015$). The campaign improved `val_bpb` from a naive warm-compile baseline of 1.0301 to a confirmed 0.9488 (+8.1%) using six levers, and then spent the majority of its effort establishing — rather than merely asserting — that this is a floor for the budget and architecture family. The trajectory has three phases. (i) Ascent: parameter-free and capacity levers (warm compile, critical batch 2^{18} , an $O(1)$ n-gram hash value-embedding, and `torch.compile max-autotune`) reach ≈ 0.9546 . (ii) Re-optimization: recognizing that a dominant structural lever (the n-gram memory) staled earlier verdicts, three iterated GPU-controlled coordinate passes both improved and simplified the model (FFN $6\times \rightarrow 4\times$, `matrix_lr` $0.04 \rightarrow 0.03$, `table` $2^{20} \rightarrow 2^{21}$), reaching a fixed point at ≈ 0.9507 . (iii) Basis expansion and floor: a new-mechanism sweep found one lever outside the tuned basis — halving the sliding-window attention span, confirmed 3/3 paired seeds, $\rightarrow 0.9488$ — after which a second, all-null sweep across seven axes (n-gram order/skip/bag, attention heads, injection point, positional, optimizer) certified a basis-limited floor, each null mechanistically at-

tributed. We distill the campaign into a lever taxonomy, an update-limited-that-is-really-epoch-limited account of the budget (the 5-minute frame sits at the ~ 2 -epoch overfitting boundary of the fixed dataset), and a ladder of nine falsifiable hypotheses (H1–H9) on autonomous search, culminating in a stopping rule: a zero-flip re-optimization pass conjoined with a mechanistically-attributed all-null new-mechanism sweep. The campaign’s central methodological result is negative and procedural: on a shared cluster near a noise floor, only GPU-controlled paired designs are admissible (one false positive, an fp8 “win,” was caught and retracted), and knowing when to stop — and proving it — is as much the work as knowing what to try.

1. Introduction

This paper is the capstone of an autonomous research campaign whose individual results appear in papers 001–009; it synthesizes the whole and states what an outside reader should take away. The task was fixed and adversarial to over-claiming: minimize validation bits-per-byte of a $\sim 50\text{M}$ -parameter nanochat-style GPT (Karpathy, 2025) trained for exactly five minutes of wall-clock on one H200 of a shared $8\times\text{H200}$ node. Every run costs the same five minutes; the seed noise floor is $\sigma \approx 0.0015$; and the shared cluster injects up to 0.0014 of `val_bpb` spread from GPU placement alone — comparable to the effect sizes that separate a good lever from a null one.

Two things make the campaign worth reporting as a whole rather than as nine separate levers. First, the shape of the trajectory: a short ascent to a plausible plateau, a long and productive re-optimization that both improved and simplified the model, a late basis-expanding discovery that reopened the search after it appeared converged, and finally a rigorous, all-null certification of the floor. Second, the discipline: near a noise floor on a contended cluster, the campaign’s

¹Autoresearch reimplement campaign. Correspondence to: Autonomous Research Agent (OPHIS) <baiyuzhu@mit.edu>.

Table 1. Terminal lever ledger (rounds 1–61, 340 runs). Adopted levers confirmed with GPU-controlled paired seeds. Baseline 1.0301; SOTA 0.9488 best single, ≈ 0.9500 pooled, +8.1%.

Lever / axis	Status	Why
Warm compile cache	+keep	kills cold-compile artifact
Batch 2^{18}	+keep	critical batch; max U
n-gram(2, 3)@ 2^{21}	+keep	O(1) memory; largest lever
FFN $6\times \rightarrow 4\times$	+keep	n-gram substitutes top slice
matrix_lr 0.03	+keep	re-optimum post-n-gram
short-window 2^9	+keep	n-gram covers local recall
table $> 2^{21}$ / order > 3	exhausted	saturated / too sparse
skip- & bag-grams	exhausted	lose adjacency / lose order
n-gram $\rightarrow$ logit head	harmful	uncontextualizable prior
heads / GQA	exhausted	VE dim load-bearing
window pattern	exhausted	span was the gain
RoPE/init/momentum/ns	exhausted	already at optimum
fp8 / sparse-grad / LS	retracted/void	artifact / slow / miscal.

dominant failure mode was not missing a lever but adopting a phantom one, and the machinery built to prevent that (GPU-controlled paired designs, a leave-one-out audit, a mechanistically-attributed stopping rule) is more transferable than any single hyperparameter.

We contribute: (1) the consolidated result and lever taxonomy (Section 2); (2) the three-phase trajectory and why re-optimization and basis-expansion were each necessary (Section 3); (3) the update-limited-that-is-epoch-limited account of the budget (Section 4); (4) the H1–H9 hypothesis ladder and the stopping rule (Section 5); and (5) the methodological lessons for autonomous research loops (Section 6).

2. The Consolidated Result

Table 1 is the campaign’s terminal ledger: six adopted levers moving val_bpb from 1.0301 to 0.9488 (+8.1%), and seven axes exhausted with identified mechanisms. The adopted levers share a signature: none adds a from-scratch-learned mapping. They are parameter-free (warm compile, max-autotune), throughput-critical (batch 2^{18}), capacity-removing (FFN $6\times \rightarrow 4\times$, short-window span $2^{10} \rightarrow 2^9$), or O(1)-lookup-addressed (the n-gram hash memory). Two of the six remove parameters while improving the metric — the update-limited regime rewards subtraction.

3. The Three-Phase Trajectory

Phase I — Ascent (rounds 1–30). Standard levers, adopted greedily, reach ≈ 0.9546 : warm compile (a 0.04 correction of a cold-cache measurement artifact, paper 001), batch 2^{18} (the critical batch, paper 001),

FFN widening (paper 002), the n-gram hash value-embedding (the single largest lever and a hash-table scaling law, paper 003 (OPHIS, 2026a)), and max-autotune compilation (paper 004). Along the way, the campaign learned its central statistical lesson the hard way: single-seed screens near $\sigma \approx 0.0015$ have a high false-flash rate, and many “wins” died under multi-seed confirmation (paper 002).

Phase II — Re-optimization (rounds 31–48). The plateau at 0.9546 was not the floor of the levers but of the search order. The n-gram memory, a dominant structural lever adopted late, had silently staled earlier verdicts: a leave-one-out audit showed FFN $6\times$ now bought nothing while still costing 16% of steps, because the n-gram memory had taken over the local-statistics work the extra width supplied (paper 007 (OPHIS, 2026c)). Reverting to FFN $4\times$ (a simplifying win) and re-sweeping flipped a second verdict (matrix_lr 0.03, previously rejected) and a third (table $2^{20} \rightarrow 2^{21}$, previously “saturated”). Three iterated GPU-controlled coordinate passes reached a fixed point at ≈ 0.9507 on a strictly smaller model (paper 008 (OPHIS, 2026d)).

Phase III — Basis expansion and the floor (rounds 49–61). A fixed point of coordinate descent is a floor only within the tuned basis. A new-mechanism sweep tested that: three axes nulled (label smoothing, n-gram order set, sparse-grad) but a fourth hit — halving the short-attention-window span, confirmed 3/3 paired seeds ($\Delta = -0.00152$), reaching the campaign best 0.9488 (paper 008). The lever is the same subtraction motif: the n-gram memory supplies local recall by table lookup, so the short layers need less span, and removing it raises U at neutral q . A fifth pass re-converged zero-flip, and a second new-mechanism sweep — window pattern, GQA/heads, skip-grams, bag-grams, a six-lever scattershot, and a direct n-gram $\rightarrow$ logit head — returned all-null, each with an identified mechanism, certifying a basis-limited floor and yielding the learning-budget constraint H9 (paper 009 (OPHIS, 2026e)).

4. Update-Limited, Really Epoch-Limited

Every phase is organized by one relation: val_bpb $\approx q \cdot U$, the product of per-update loss q and update count U (papers 001, 004). A lever helps only by lowering q without proportionally lowering U , or raising U at fixed q ; this is why capacity-removing and O(1) levers dominate and why throughput matters as much as modeling.

A late meta-experiment (round 61) sharpened “update-limited” into “epoch-limited.” Doubling the budget to ten minutes did not lower `val_bpb`; it raised it ($0.9488 \rightarrow 0.987$), because the fixed training set is small enough that five minutes is already ≈ 2 epochs and ten minutes is three — the model overfits on repeated data. The 5-minute frame therefore sits at the ~ 2 -epoch overfitting boundary, which is precisely why U cannot be freely increased and why the regime is update-limited at all. It also makes cross-budget lever comparison confounded on fixed data — at ten minutes a slower configuration (FFN6 $\times$) “wins” only by completing fewer epochs and overfitting less — so H9’s budget-dependence clause is untestable here without a larger corpus. We report this as a confound honestly rather than as a budget-scaling result.

5. The Hypothesis Ladder (H1–H9)

The campaign’s papers each posed a falsifiable hypothesis; together they form a ladder from measurement hygiene to a stopping rule.

- H1–H4 (papers 001–004): measurement artifacts (cold-compile cache) masquerade as levers; single-seed screens near the floor are mostly false; the n -gram hash-table follows a scaling law that saturates; the step is bounded by small-GEMM/fill throughput, not attention.
- H5–H6 (papers 005–006): the campaign is update-limited ($q \cdot U$); and on a shared cluster, unpaired A/B is confounded by GPU contention — an fp8 “win” was a placement artifact and was retracted.
- H7 (paper 007): greedy sequential adoption leaves levers stale; a dominant structural adoption is a trigger to re-optimize, which typically improves and simplifies.
- H8 (paper 008): iterated re-optimization converges to a stack-level fixed point; a new-mechanism sweep is then dual-purpose — an all-null sweep certifies a basis-limited floor, a hit expands the basis and triggers one more pass.
- H9 (paper 009): under a fixed update-limited budget, a new mechanism helps only if usable without learning its addressing from scratch — $O(1)$ lookup memories qualify, learn-to-address memories do not (an $O(1)$ n -gram hash is the largest lever; a learnable n -gram $\rightarrow$ logit head is catastrophic).

The operational distillate is a stopping rule: declare a basis-limited floor when a full re-optimization pass is zero-flip and a new-mechanism sweep returns only nulls, each attributable to an identified cause rather than to noise. Both conjuncts are necessary — without the sweep, a zero-flip pass may reflect an impoverished proposal set; without the fixed point, a null sweep may be swamped by an un-re-optimized stack.

6. Methodological Lessons for Autonomous Loops

Paired designs are not optional. The retracted fp8 result (paper 006 (OPHIS, 2026b)) is the campaign’s most important negative: a $+0.001$ “win” that was GPU-placement noise. Thereafter every comparison interleaved and GPU-swapped arms; a same-config control spread of 0.0014 makes unpaired comparison uninterpretable at this resolution. An autonomous loop on shared hardware must build contention control in, not bolt it on.

Verdicts are conditional; re-optimize after structural change. Half the campaign’s late gains came not from new levers but from re-testing old verdicts after a structural adoption moved their optima (H7). A cheap leave-one-out audit ($k+1$ runs) measures each lever’s current marginal value and directly exposes stale ones.

Stopping is a result. The campaign spent Phase III proving a floor, not chasing it. A mechanistically-attributed all-null sweep is the responsible terminal act of a loop on a shared cluster: it distinguishes “converged” from “out of ideas,” and it tells the operator that further gains need a new budget, corpus, or architecture family — not more within-basis search.

Prefer subtraction and lookup. Four of the campaign’s wins remove parameters or are parameter-free; the largest modeling lever is a fixed hash lookup; the clearest failure (round 58) added a from-scratch-learned map on the logits. Under a tight budget, the admissible-mechanism set is dominated by parameter-free, capacity-removing, and $O(1)$ -addressed structure (H9).

7. Limitations and Next Directions

The floor is certified for one budget, one ~ 50 M architecture family, and one fixed corpus; H9’s budget clause is untestable here (Section 4) and is the first thing a larger-corpus follow-up should settle. The most promising untried directions, ranked: (1) test H9 directly with a learned k -NN/product-key memory

(Lample et al., 2019), predicted to fail at 5 minutes but clear the gate at $2\text{--}4\times$ budget with fresh data; (2) a parameter-free longer-range ordered hash feature (baggrams failed by discarding order, round 59) that stays dense without losing sequence; (3) order-independence of the fixed point (H7 Direction 3), adopting the same levers in a different sequence and checking the terminal stack is invariant; (4) automating the H8+H9 stopping rule as a proposal pre-filter, which would have rejected the catastrophic logit head a priori. Each is stated with a falsifier in the source papers.

8. Conclusion

Over 61 rounds an autonomous agent took a fixed-budget nanochat GPT from 1.0301 to 0.9488 val_bpb (+8.1%) with six levers — most of them subtractive or lookup-based — and then did the harder thing: it proved, mechanism by mechanism, that this is a floor for the budget and architecture, retracting one false positive and stopping on a certified all-null sweep rather than churning the shared cluster. The transferable content is not the levers but the loop: control for contention with paired designs, re-optimize after every structural change because verdicts are conditional, prefer mechanisms that need no from-scratch learning under a tight budget, and treat a mechanistically-attributed stopping rule as a first-class result. An autonomous research loop earns its keep not only by finding improvements but by knowing — and proving — when it has found the floor.

References

- Karpathy, A. nanochat: The best ChatGPT that \$100 can buy. 2025. URL <https://github.com/karpathy/nanochat>.
- Lample, G., Sablayrolles, A., Ranzato, M., Denoyer, L., and Jégou, H. Large memory layers with product keys. NeurIPS, 2019.
- OPHIS Autonomous Research Agent. N-gram hash value-embeddings and a hash-table scaling law in fixed-wall-clock LLM pretraining. Campaign paper 003, AI_papers/, 2026a.
- OPHIS Autonomous Research Agent. Throughput levers and a retracted false positive: GPU-contention confounds in shared-cluster A/B testing. Campaign paper 006, AI_papers/, 2026b.
- OPHIS Autonomous Research Agent. Lever staleness under greedy sequential adoption. Campaign paper 007, AI_papers/, 2026c.

OPHIS Autonomous Research Agent. Convergence of iterated re-optimization and a mechanism-exhaustion map. Campaign paper 008, AI_papers/, 2026d.

OPHIS Autonomous Research Agent. The basis-limited floor and a learning-budget constraint on new mechanisms. Campaign paper 009, AI_papers/, 2026e.